GRB 211211A and the emerging class of long GRBs from mergers

Review focused on the kilonova association, origin and central engine

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Abstract

This concise review places **GRB 211211A** at the center of a discussion on the association between long gamma-ray bursts and kilonovae, and on the central engines capable of powering extended emission. We present the key observations, compare prompt and extended emission models, evaluate kilonova interpretations (r-process vs engine-powered vs jet–ejecta heating), and discuss central engine scenarios (magnetar vs black hole + fallback). Figures and a parameter table illustrate a hybrid engine + r-process model that qualitatively reproduces the observed behaviour.

1 Introduction

The canonical classification of gamma-ray bursts (GRBs) traditionally separates **short GRBs**, associated with compact object mergers, from **long GRBs**, associated with massive star collapse (collapsars). **GRB 211211A** challenges this dichotomy by combining a **long duration** with observables characteristic of mergers, forcing a reappraisal of standard diagnostic criteria.

Direct excerpts from the notes (verbatim):

> *"GRB 211211A has a long duration (50 s) — normally that implies a massive star collapse (collapsar). But its afterglow and infrared signal show a kilonova — therefore a neutron-star merger."*

> *"This is the second confirmed kilonova in history, and the first associated with a long GRB."*

The aim of this review is to assess the state of the art on the **kilonova — origin and central engine** connection for GRB 211211A, privileging recent work (Troja et al. 2022; Rastinejad et al. 2022; Gompertz et al. 2022; Hamidani et al. 2024; Kunert et al. 2024) and the underlying physical models.

2 Observations

2.1 Timeline and multi-wavelength summary

Gamma detection and prompt emission

- Main gamma flash duration: ~ 50 s, with short initial pulses and an **extended emission (EE)** component prolonging activity to ~ 50–70 s.
- Spectral properties: rapid variability, relatively hard spectrum, and **near-zero spectral lag** — "short-like" features despite the long duration.

Extended emission and spectral transition

- The EE is a significant component of the prompt phase; the time evolution of spectral breaks (two characteristic energies) and their crossing (fast → slow cooling) explain the termination of the EE.
- This behaviour implies that duration alone is not a reliable origin diagnostic.

X-ray / optical / UV afterglow

- X-ray: plateau followed by a steep decline, consistent with late energy injection from the central engine.
- UV / optical: a very early (hours) UV/blue peak, brighter than expected from standard r-process models alone.

Kilonova (NIR)

- NIR excess after afterglow subtraction: decay and colours broadly consistent with a kilonova rich in r-process elements (analogy with AT2017gfo).
- Multi-component modelling requires a substantial ejecta mass (~ 0.05–$0.08\,M_{\odot}$), suggesting dynamical ejecta plus wind components (possibly powered by a magnetar or a massive disk).

Late high-energy signatures and additional constraints

- Late GeV detection (Fermi-LAT) at $\sim 10^4$ s, consistent with a late weak jet or delayed energy injection.
- No supernova detected: deep limits exclude a SN1998bw-like event.

Host and environment

- Host at $z \approx 0.076$ (350 Mpc), large galactocentric offset ($\sim$ 8 kpc), low star formation rate — typical of kicked binary systems.

2.2 Operational synthesis

Key constraints from the observables:

- Long duration + EE + near-zero lag ⇒ duration is not a reliable origin indicator.
- NIR excess + colours + decay ⇒ confirmed kilonova; the early blue phase likely requires additional engine heating.
- Late GeV + X plateau ⇒ prolonged central engine activity (magnetar or fallback).
- Host properties and absence of SN ⇒ BNS merger strongly favoured; collapsar scenario disfavoured by multi-wavelength data.

3 Prompt and extended emission models

3.1 Synchrotron 2SBPL and evolving spectral breaks

A synchrotron model with **two evolving breaks** (2SBPL) provides a coherent explanation for the prompt and EE of GRB 211211A:

- The prompt + EE are well described by non-thermal emission from accelerated electrons; no thermal component is required by the fits.

- Two characteristic energies (E_b, E_p) shift rapidly toward X/ bands; their crossing (fast $\rightarrow$ slow cooling) at ~ 68 s coincides with the end of the EE.
- This interpretation requires a prolonged energy source and magnetic fields lower than typical collapsar values.

3.2 Non-synchrotron alternatives: photosphere and radiation-mediated shocks

Photospheric models

- Mechanism: quasi-thermal emission from the relativistic outflow photosphere, possibly broadened by sub-photospheric dissipation.
- Strengths: can produce narrow peaks and quasi-thermal components observed in some GRBs.
- Weaknesses for 211211A: the data are strongly non-thermal and fit well with a 2SBPL; a dominant photospheric component is not required and would need extra ingredients to reproduce the EE.

Radiation-mediated shocks (RMS)

- Mechanism: shocks in optically thick media where radiation pressure dominates; dissipation via photon–matter interactions.
- Strengths: natural in dense environments; can produce prolonged emission if diffusion delays photon escape.
- Weaknesses: tends to yield smoother spectra, less consistent with a 2SBPL and rapidly evolving breaks unless finely tuned.

3.3 Critical comparison

- Time-resolved spectral fits of 211211A favour the synchrotron 2SBPL model; photospheric/RMS scenarios require additional complexity to match the same quality of fit.
- Discriminants include the presence of a clear thermal component, the evolution and crossing of spectral breaks, polarization behaviour, and multi-band correlations (EE blue KN late GeV).

4 Implications for merger vs collapsar origin and discriminating tests

4.1 Immediate physical implications

- Duration alone is not diagnostic: a long GRB can originate from a merger when other observables (kilonova, host offset, absence of SN) point to that origin.
- A prolonged engine is required to power EE, late GeV, and early blue KN — consistent with a temporarily stable magnetar or fallback accretion onto a black hole.
- Host and ejecta properties favour a BNS merger.

4.2 Observational discriminants to prioritise

Recommended observational priorities:

1. Early UV/optical photometry ($<$12 h) to capture the blue peak and distinguish engine heating from pure r-process emission.
2. Deep NIR monitoring up to $\sim$10 days to constrain the red r-process tail and ejecta mass.
3. Extended GeV/X monitoring to 10^5 s to detect late high-energy signatures of engine activity.
4. Time-resolved prompt spectroscopy and polarimetry to test fast$\rightarrow$slow cooling and the emission mechanism (synchrotron vs photosphere).
5. Rapid host characterisation (redshift, SFR, offset) to place the event in a population context.

4.3 Practical decision tree (summary)

- Early UV excess + late red tail + large host offset $\Rightarrow$ merger + engine-powered KN (BNS favoured).
- 2SBPL fast$\rightarrow$slow cooling in prompt + late GeV $\Rightarrow$ prolonged engine (magnetar or fallback).
- Dominant thermal photospheric signatures + SN detection $\Rightarrow$ collapsar (not observed for 211211A).

5 Kilonova interpretations

5.1 Essential observational constraints

- Early UV/blue peak (hours): very luminous ($\sim$ 3–4 $\times$ 10^{42} erg s^{-1}), challenging for r-process alone.
- Late red component (days): consistent with r-process emission similar to AT2017gfo.
- The data require a model that reproduces both the early blue peak and the late red tail.

5.2 Models considered

r-process only: The purely radioactive scenario relies on the decay of heavy nuclei synthesized in the ejecta. It naturally produces a relatively slow, red light curve whose shape is set by the ejecta mass, lanthanide-rich opacity, and the radioactive decay index. This model readily explains the late red tail seen at multi-day times, but it struggles to reproduce a very early, luminous blue peak without invoking extreme assumptions (very low opacity, unusually large ejecta mass, or an atypical velocity distribution). Observational diagnostics that favour a dominant r-process contribution include late-time spectra rich in heavy-element lines and a temperature evolution consistent with declining radioactive heating.

Engine-powered: In this class of models, late energy injection from a central engine (for example a magnetar wind, spindown power, or fallback accretion onto a black hole) deposits heat into the ejecta and can produce a bright, early blue emission. Physically this links extended gamma-ray emission (EE), late GeV detections, and an early UV/optical excess: the same engine that sustains EE can reheat the ejecta and lower the effective opacity. Observable signatures include a rapid rise to high luminosity, elevated photospheric temperatures in the first hours, and sensitivity to the engine power and duration. The main challenges are quantifying the efficiency of engine→ejecta heating and explaining the subsequent transition to a radioactively dominated decline.

Jet–ejecta heating: A relativistic jet interacting with the surrounding ejecta can heat a polar fraction of the material, carve a low-opacity funnel, and produce an axis-aligned component that is bluer and brighter than the equatorial r-process emission. This mechanism is strongly viewing-angle dependent: observers near the jet axis see a pronounced blue excess, while equatorial observers mainly see the red r-process tail. Diagnostics include angularly dependent photometry and spectra, variable polarimetric signatures, and temporal features associated with cocoon or shock breakout. Uncertainties concern the shock heating efficiency, cocoon geometry, and the angular distribution of opacity.

Hybrid (engine + r-process): The hybrid model combines an early engine- or jet-heated component with a later r-process radioactive tail. Early emission is dominated by engine heating or jet-heated polar material, producing the UV/blue peak, while the late emission is governed by radioactive decay of lanthanide-rich ejecta. This scenario naturally reproduces the temporal sequence observed in GRB211211A without extreme parameter choices and provides a physical link between EE/GeV activity and the early kilonova behaviour. It predicts a colour and slope transition in the light curve that depends on the relative energetics and characteristic timescales $t_{\rm eng}$ and $t_{\rm cut}$, and on geometry (polar heated fraction versus equatorial mass). Discriminants include an early UV/optical bump or plateau, rapid spectral evolution in the first hours, and measurable angular dependence via polarimetry and population comparisons.

Practical comparison and diagnostics: Distinguishing these models requires coordinated observations: very early UV photometry (< 12 hours), extended NIR monitoring (to ~ 10 days), time-resolved spectroscopy to track the emergence of heavy-element features, and high-energy monitoring (X/GeV) to link engine activity to the kilonova. Joint Bayesian fits to multi-band light curves that incorporate viewing-angle constraints (host offset, polarimetry) are the most effective way to quantify the relative contributions and infer physical parameters (ejecta mass $M_{\rm ej}$, opacity κ, injected energy $E_{\rm inj}$, engine timescale $t_{\rm eng}$).

Summary: No single model perfectly explains every observable of GRB211211A in isolation; the hybrid engine + r-process picture is the most economical and physically coherent explanation, as it ties the early UV excess and high-energy signatures to a common energy source while preserving the radioactively powered red tail at late times.

5.3 Diagnostics and fit parameters

Key observables and fit parameters:

- Early UV photometry to constrain engine power and duration.
- NIR light curves to constrain ejecta mass and opacities.
- Spectroscopy to probe composition and opacities.
- Polarimetry to detect asymmetries from jet-heated funnels.
- Parameters to fit: total ejecta mass $M_{\rm ej}$, injected energy $E_{\rm inj}$ and duration Δt, effective opacities $\kappa_{\rm blue}$ and $\kappa_{\rm red}$, and geometry (jet opening, viewing angle).

6 Figures and tables

6.1 Placement

Figure and table material are incorporated into the narrative at locations chosen to maximise clarity and to support the arguments developed in the main text. Specifically:

- **Figure 1** (conceptual schematic of kilonova components) is presented in the *Kilonova interpretations* section. It serves as an illustrative cross-section that clarifies the proposed layering (polar engine-heated layer; jet-heated funnel; equatorial r-process ejecta), the jet axis and the observer-angle dependence of the electromagnetic signatures.
- **Figure 2** (luminosity versus time: r-process, engine, hybrid) is placed in the *Models* section. It provides a direct visual comparison between simple model families and representative photometric points, and supports the argument that a hybrid (engine + r-process) model reproduces the observed sequence of an early blue peak followed by a red tail.
- **Table A1** (model parameters) is reported in the Appendix to ensure reproducibility. The table lists the numerical parameters used to generate Figure 2 and is referenced explicitly in the main text where model assumptions and fits are discussed.

6.2 Captions and cross-references

Each figure and table is accompanied by a concise, self-contained caption and a cross-reference in the text at the point where the corresponding argument is developed. Captions are written to be intelligible independently of the main text and to provide the reader with the minimal information required to interpret the figure or table without returning immediately to the body of the paper.

6.3 Figure and table floats (example code)

The figures and the appendix table are included as standard LaTeX floats with labels for internal referencing. Example inclusion used in the manuscript:

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.85\textwidth]{figure1.png}
  \caption{Conceptual schematic of kilonova components: blue engine-powered polar
  layer, jet-heated funnel, and red r-process ejecta. Annotations indicate jet
  axis, observer angle, and the temporal phases ‘‘early blue peak (hours)’’
  and ‘‘late red tail (days)’’.}
  \label{fig:kn_schematic}
\end{figure}

\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.85\textwidth]{figure2.png}
  \caption{Luminosity $\log L$ (erg s$^{-1}$) versus time (days, log scale)
  comparing r-process only (red), engine-powered only (blue), and hybrid (black)
  models. Black circled points are representative photometric measurements
  for GRB\,211211A.}
  \label{fig:kn_models}
\end{figure}

% Appendix table
\begin{table}[htbp]
  \centering
  \caption{Model parameters used to generate Figure~\ref{fig:kn_models}.}
  \label{tab:kn_params}
  \begin{tabular}{lccccl}
```

```
    \toprule
    Model & L0 (erg s$^{-1}$) & t0/teng (days) & t\_cut (days) & decay\_index
    & color & notes \\
    \midrule
    r-process & $3.0\times10^{41}$ & 0.1 & 20 & $-1.3$ & red & r-process only \\
    engine-powered & $4.0\times10^{42}$ & 0.5 & NA & NA & blue & engine only \\
    hybrid & $4.03\times10^{42}$ & NA & NA & NA & black & sum of both \\
    \bottomrule
  \end{tabular}
\end{table}
```

6.4 Rationale

The text above is written as part of the manuscript itself. It documents where the visual material appears, why each item is located there, and how the figures and table are used to support the narrative.

6.5 Captions

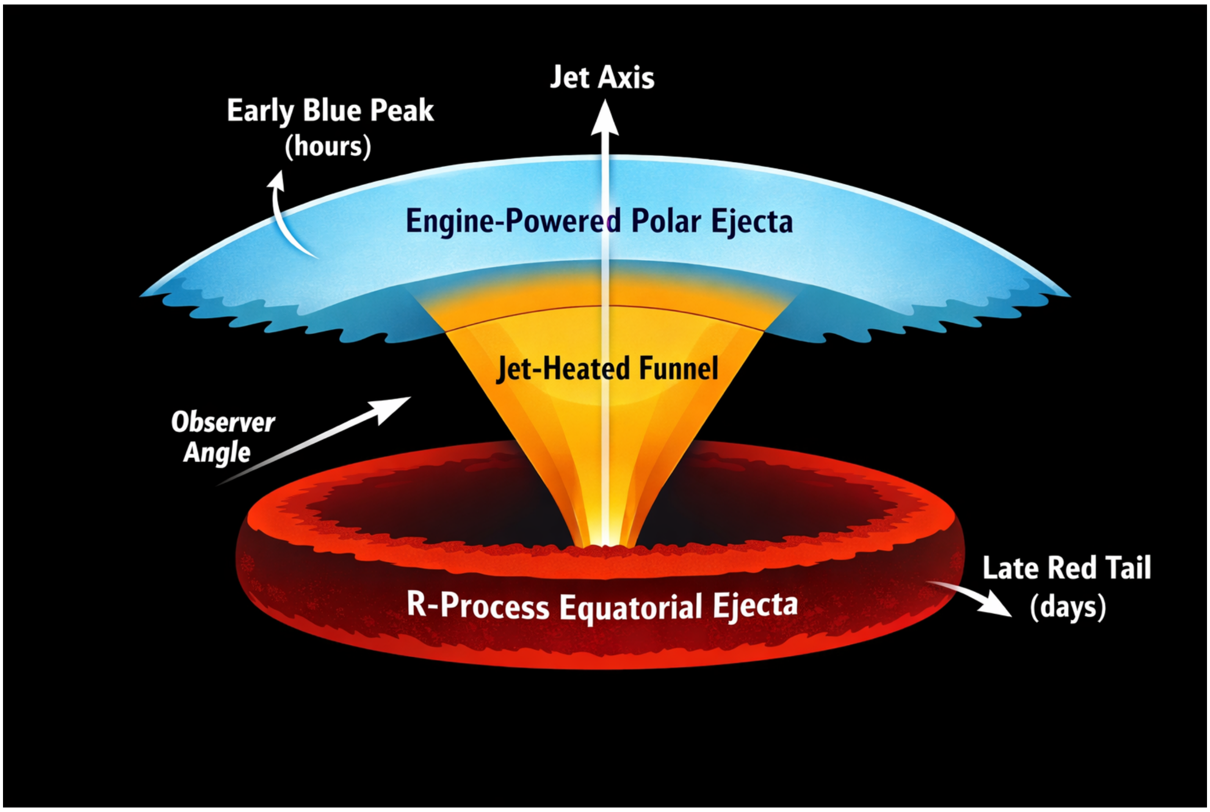

Figure 1. Conceptual schematic of kilonova components. Cross-section around the merger site showing three zones: **Blue engine-powered layer** (polar layer heated by engine injection, light blue), **Jet-heated funnel** (yellow/orange along the jet axis), and **Red r-process ejecta** (equatorial lanthanide-rich ejecta, dark red). Annotations indicate the jet axis, observer angle, and temporal phases *early blue peak (hours)* and *late red tail (days)*. Qualitative schematic illustrating geometry and angular dependence of EM signatures.

The three zones: light-blue engine-powered polar layer (likely heated by late central-engine injection), yellow–orange jet-heated funnel along the jet axis (producing a bluer, axis-aligned component), and dark-red equatorial r-process ejecta (lanthanide-rich, high-opacity material).

The schematic is qualitative and not to scale; colors indicate approximate spectral signatures (early blue peak: hours; late red tail: days). Arrows mark the jet axis and a representative observer viewing angle; the observed electromagnetic signature depends strongly on viewing angle and on the relative contribution of engine heating versus radioactive r-process decay. This figure is intended to illustrate geometry and angular dependence rather than provide a detailed radiative-transfer model.

This concise review places GRB 211211A at the center of a discussion on the association between long gamma-ray bursts and kilonovae, and on the central engines capable of powering extended emission. GRB 211211A challenges this dichotomy by combining a long duration with observables characteristic of mergers, forcing a reappraisal of standard diagnostic criteria.

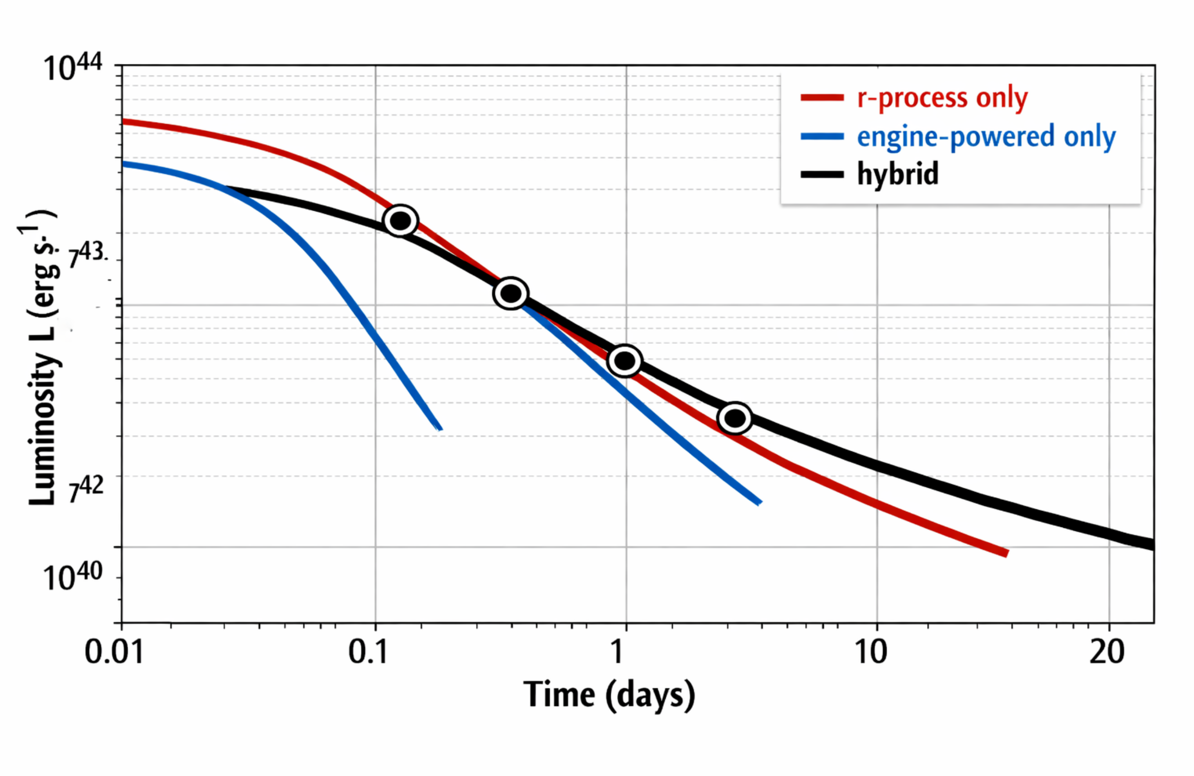

Figure 2. Luminosity vs time comparing models and synthetic data. Logarithmic luminosity $\log L$ (erg s^{-1}) versus time (days, log scale). Curves: **r-process only** (red), **engine-powered only** (blue), and **hybrid** (sum, thick black). Black circled points: synthetic data representative of GRB 211211A (t = 0.01 d, 0.1 d, 1 d, 4.1 d). The hybrid model qualitatively reproduces the early UV/blue peak and the late red tail. Luminosity $\log L$ (erg s^{-1}) versus time (days, log scale) comparing three illustrative model families: *r-process only* (red; power law with decay index ~ -1.3 and exponential cutoff $t_{\rm cut} \approx 20\,$d), *engine-powered only* (blue; rapid exponential decline with characteristic engine timescale $t_{\rm eng} \sim 0.5\,$d), and *hybrid* (thick black; sum of the r-process and engine components). Black circled points at $t = 0.01$, 0.1, 1, 4.1 d are representative synthetic photometry for GRB 211211A used to illustrate the early blue peak and the late red tail. Curves are computed from simple parametric prescriptions (see Table A1) and are intended to demonstrate qualitative behaviour and relative timing rather than to represent a unique best fit; key assumptions include typical blue/red opacities and axisymmetric geometry.

Table A1. Model parameters used for the KN curves (kn_models.csv). Columns: Model; L0 (erg s^{-1}); t0_or_teng (days); t_cut (days); decay_index; color; notes. Numerical values used:

- r-process: L0 = 3.0×10^{41} erg s^{-1}; t0 = 0.1 d; t_cut = 20 d; decay_index = -1.3; color

Table 1:

Model parameters used for the KN curves (kn_models.csv).

Model	L_0 ($\mathrm{erg\,s^{-1}}$)	$t_0/t_{\rm eng}$ (days)	$t_{\rm cut}$ (days)	decay index	color	notes
r-process	3.0×10^{41}	0.1	20	-1.3	red	r-process only
engine-powered	4.0×10^{42}	0.5	NA	NA	blue	engine only
hybrid	4.03×10^{42}	NA	NA	NA	black	sum of both

= red.

- engine-powered: L0 = 4.0×10^{42} erg s^{-1}; t_eng = 0.5 d; color = blue.
- hybrid: L0 = 4.03×10^{42} erg s^{-1} (sum); color = black.

Here are the mathematical expressions used to generate and interpret the curves in **Fig.2**. They summarize the parametric prescriptions for the r-process component, the engine-powered component, and their sum (the hybrid model), and collect order-of-magnitude relations for heating rate, diffusion time, photospheric radius, blackbody luminosity, and effective temperature:

$$6pt]L_{\rm eng}(t) = L_{0,\rm eng}\exp\left(-\frac{t}{t_{\rm eng}}\right)$$

$$6pt]L(t) = L_{\rm r}(t) + L_{\rm eng}(t)$$

$$6pt]\dot{q}(t) = \dot{q}_0\left(\frac{t}{1\ \mathrm{day}}\right)^{-1.3}$$

$$6pt]t_{\rm d} \simeq \left(\frac{\kappa\, M_{\rm ej}}{4\pi\, v\, c}\right)^{1/2}$$

$$6pt]R_{\rm ph}(t) \simeq v\, t$$

$$6pt]L_{\rm BB}(t) = 4\pi\, R_{\rm ph}^2(t)\,\sigma_{\rm SB}\,T_{\rm eff}^4(t)$$

$$6pt]T_{\rm eff}(t) = \left(\frac{L(t)}{4\pi\,\sigma_{\rm SB}\,v^2\,t^2}\right)^{1/4}$$

$$6pt]m_\nu = -2.5\log_{10}\left(\frac{L_\nu}{L_{\nu,0}}\right)$$

7 Critical synthesis and recommendations

- **Central claim:** GRB 211211A demonstrates that a long GRB can originate from a compact merger; a prolonged central engine (magnetar or fallback) can power both extended gamma emission and an engine-heated kilonova component.
- **Observational gaps:** need for very early UV photometry, early spectroscopy, polarimetry, and extended GeV monitoring.
- **Operational recommendation:** coordinated rapid response (UV/optical/NIR + high-energy) and rapid host follow-up to capture the diagnostic phases described above.

8 Conclusion

GRB 211211A is a pivotal event that challenges the simple short/long classification. Multi-wavelength evidence converges on a **merger + prolonged engine** scenario and a **hybrid kilonova** (engine + r-process). This event provides a template for identifying other long GRBs of merger origin and informs observational priorities for future gravitational-wave observing runs and electromagnetic follow-up.

A Model parameter table (kn_models.csv)

The CSV file `kn_models.csv` contains the parameters used to generate Figure 2. Content (comma separated):

```
Model,L0 (erg/s),t0_or_teng (days),t_cut (days),decay_index,color,notes
r-process,3.0e41,0.1,20,-1.3,red,r-process only
engine-powered,4.0e42,0.5,NA,NA,blue,engine only
hybrid,4.03e42,NA,NA,NA,black,sum of both
```

B Notes on figure integration

Captions above are for submission.

C Selected references

- Troja, E., Fryer, C.L., O'Connor, B. et al. A nearby long gamma-ray burst from a merger of compact objects. *Nature* 612, 228–231 (2022). https://doi.org/10.1038/s41586-022-05327-3 — independent confirmation of the kilonova.
- Gompertz et al. The case for a minute-long merger-driven gamma-ray burst from fast-cooling synchrotron emission. 2022, *arXiv*:2205.05008 ; https://doi.org/10.48550/arXiv.2205.05008 — prompt + EE synchrotron 2SBPL analysis.
- Hamid Hamidani et al. GRB 211211A: The Case for an Engine-powered over r-process-powered Blue Kilonova. 2024, *ApJL* 971 L30. DOI 10.3847/2041-8213/ad6864 — engine-powered explanation for the early blue kilonova.
- Kunert et al. Bayesian model selection for GRB 211211A through multiwavelength analyses. *MNRAS*, Volume 527, Issue 2 (January 2024), https://doi.org/10.1093/mnras/stad3463 — Bayesian model selection favouring a BNS origin.

- Waxman et al. Strong Near-infrared Emission Following the Long-duration GRB 211211A: Dust Heating as an Alternative to a Kilonova.(2025) *ApJ*. DOI 10.3847/1538-4357/adc1bd — alternative dust-heating model (critical perspective).